

# **INTERNSHIP PROJECT REPORT**

## **Fake News Detection System using Machine Learning**

### **1. Introduction**

With the rapid growth of digital media and social networking platforms, the spread of fake news has become a serious global issue. Misinformation can influence public opinion, create panic, and affect political and social stability. Detecting fake news manually is difficult due to the massive volume of content published daily.

This project aims to develop an automated Fake News Detection system using Machine Learning and Natural Language Processing (NLP) techniques to classify news articles as either Real or Fake.

### **2. Problem Statement**

The objective of this project is to build a binary classification model that can accurately distinguish between real and fake news articles based on textual content. The system should:

- Process textual news data
- Extract meaningful features
- Train a classification model
- Predict whether a given news article is real or fake
- Provide confidence score for predictions

### **3. Dataset Description**

The dataset used consists of two CSV files:

- True.csv – Contains real news articles
- Fake.csv – Contains fake news articles

Each dataset includes:

- Title of the news article
- News content (text)
- Subject
- Date

After loading both datasets:

- Real news articles were labeled as 1

- Fake news articles were labeled as 0
- Both datasets were merged into a single dataset for training.

## **4. Tools and Technologies Used**

- Python – Programming language
- Pandas – Data manipulation
- NLTK – Text preprocessing
- Scikit-learn – Machine Learning model
- TF-IDF Vectorizer – Feature extraction
- Logistic Regression – Classification algorithm
- Streamlit – Web application development
- VS Code – Development environment

## **5. Methodology**

### **5.1 Data Preprocessing**

Text data was cleaned using the following steps:

- Conversion to lowercase
- Removal of special characters and numbers
- Removal of stopwords using NLTK
- Combination of title and content into a single feature

This ensures noise-free input for the machine learning model.

### **5.2 Feature Extraction**

Text data cannot be directly used in machine learning models. Therefore, TF-IDF (Term Frequency – Inverse Document Frequency) was used to convert text into numerical feature vectors.

Improvements made:

- Maximum features set to 10,000
- N-grams range set to (1,2)
- Stop words automatically removed

TF-IDF helps in identifying important words in a document relative to the dataset.

### 5.3 Model Training

The dataset was split into:

- 80% Training Data
- 20% Testing Data

A Logistic Regression classifier was used because:

- It performs well for binary classification
- Efficient with high-dimensional sparse data
- Fast and interpretable

Class balancing was applied using `class_weight="balanced"` to handle potential imbalance.

### 5.4 Model Evaluation

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score

The model achieved approximately:

Accuracy: 99%

This indicates strong performance on unseen test data.

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## 6. Deployment

The trained model was saved using Pickle and integrated into a web application using Streamlit.

The web application allows users to:

- Enter a news article
- Get prediction (Real / Fake)
- View confidence percentage

This makes the system interactive and user-friendly.

## **7. Results**

The developed Fake News Detection system successfully classifies news articles with high accuracy and stability.

The use of TF-IDF with Logistic Regression proved effective for text-based classification tasks.

The web application demonstrates practical deployment of machine learning in real-world scenarios.

## **8. Conclusion**

This project demonstrates the application of Natural Language Processing and Machine Learning techniques to detect fake news effectively. The system achieved high accuracy and provides real-time predictions through a web interface.

Future improvements may include:

- Using Deep Learning models (LSTM, BERT)
- Expanding dataset for better generalization
- Multi-class classification
- Integration with social media APIs

The project successfully meets its objective and provides a scalable solution for misinformation detection.